

01 Detect

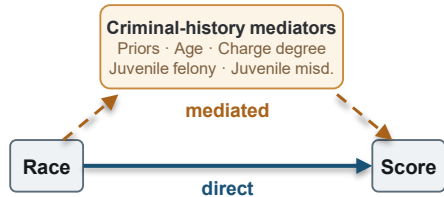
Correlation baseline

✓ DIR 1.74 · 4/5 rule fails

02 Discover

Causal structure search

6 algorithms · 4 families



03 Isolate

Direct vs. proxy pathway

✓ Where the fix is decided

► DIRECT PATHWAY

Race → **Score**

Discriminatory bias embedded.

Intervention: revise internal logic.

► PROXY PATHWAY

Race → **Priors** → **Score**

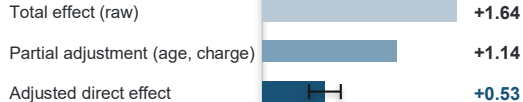
Structural inequities inflate a “neutral” risk signal. **Intervention: upstream policy reform -removing Race will not eliminate this path.**

04 Quantify

The Causal Effect. Apply covariate adjustment to estimate the total and adjusted direct effect.

✓ +1.64 attenuates to +0.53

EFFECT ATTENUATION UNDER CONTROLS



05 Validate

Bootstrap resampling

✓ 1,000 bootstrap draws
✓ CI excludes 0
✓ 95% CI [+0.40, +0.66] ·
✓ E-value 1.65